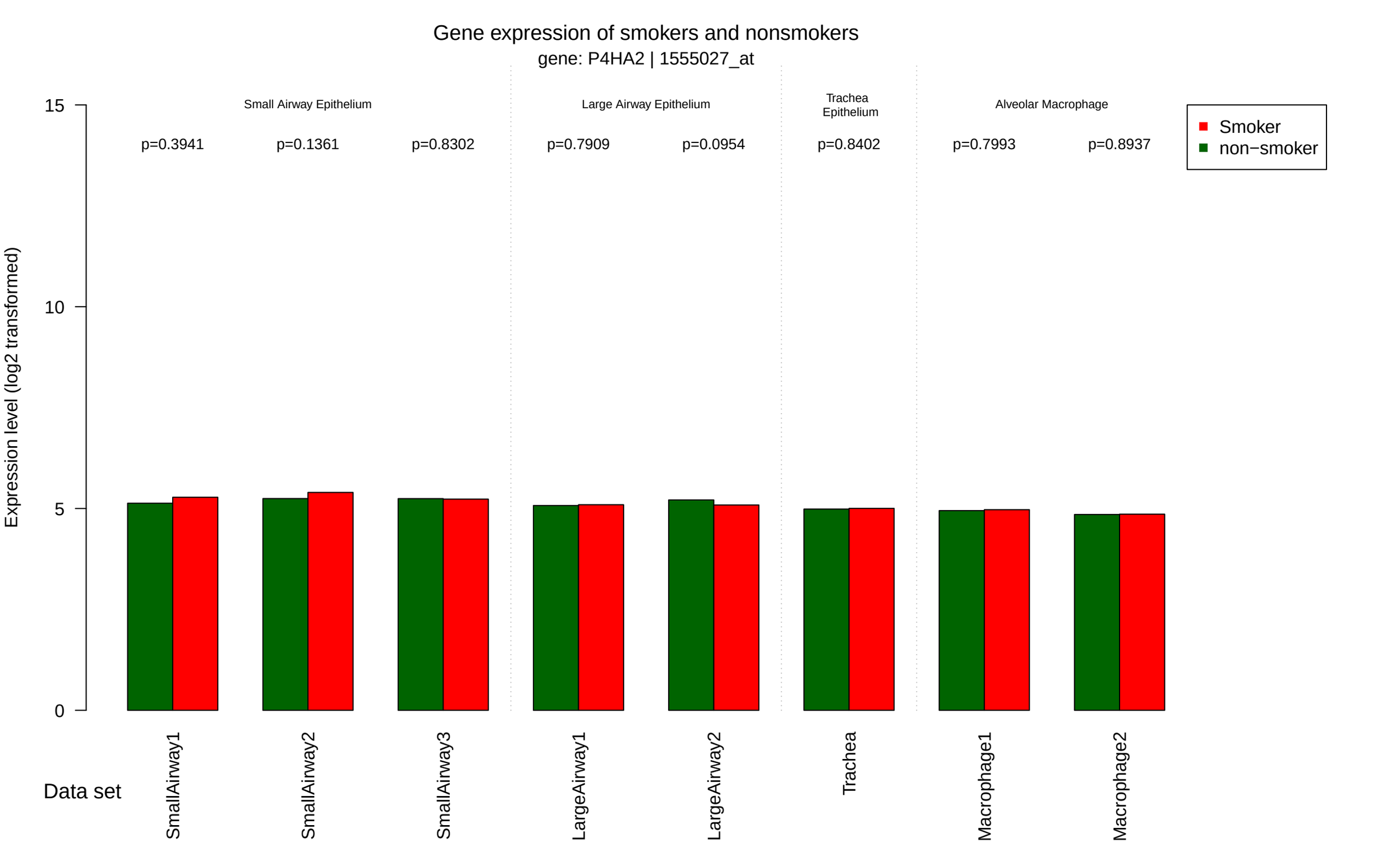
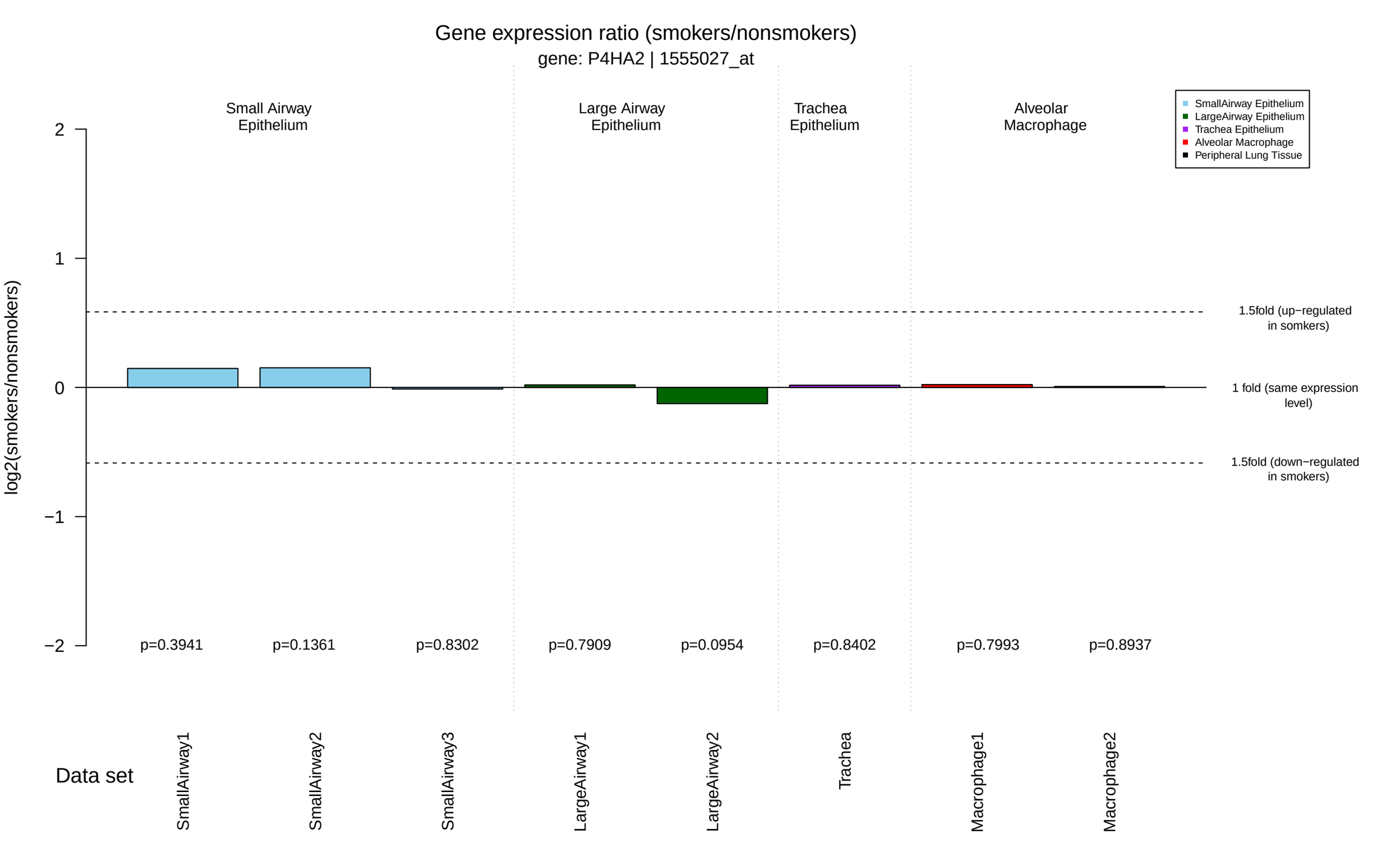
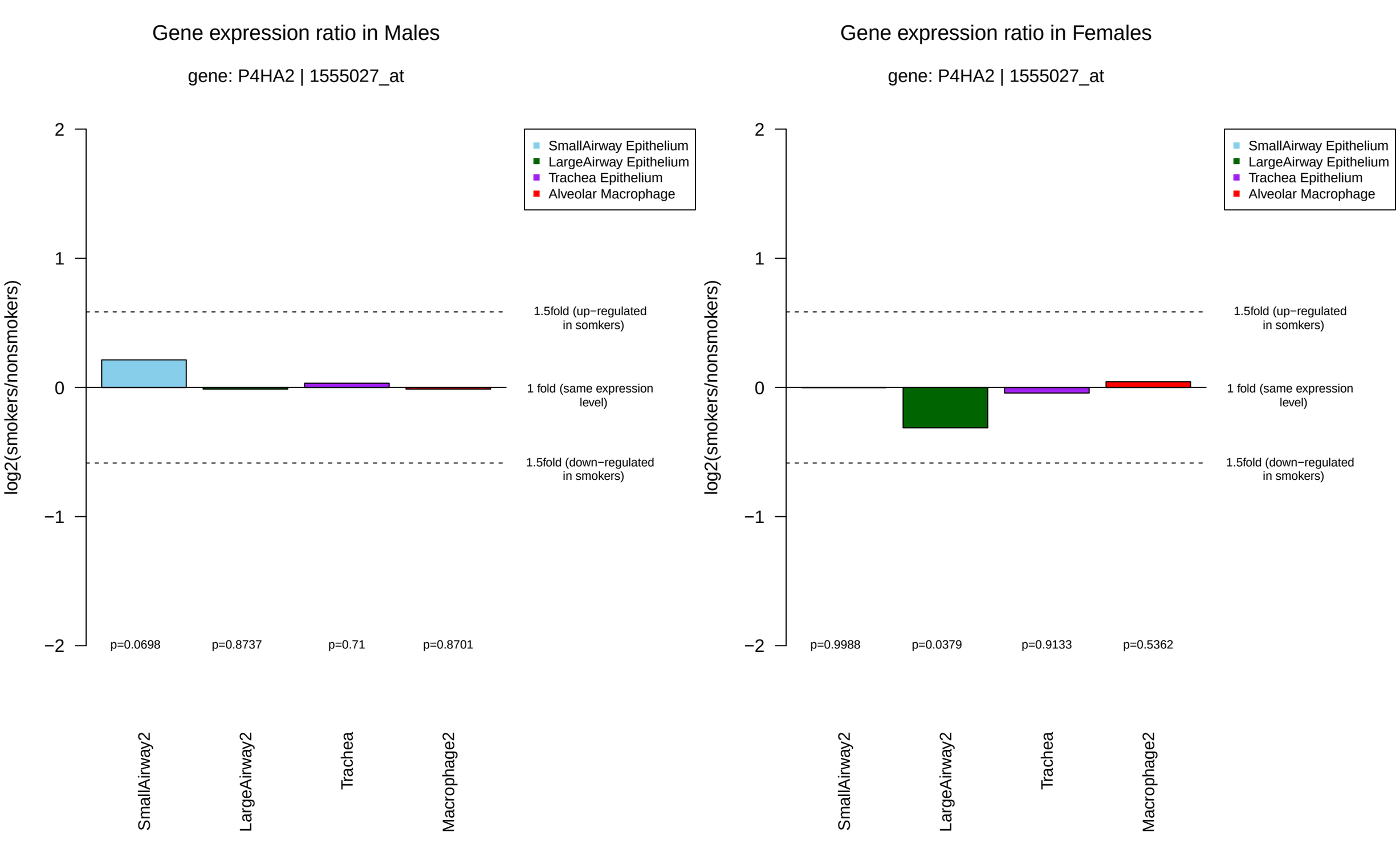
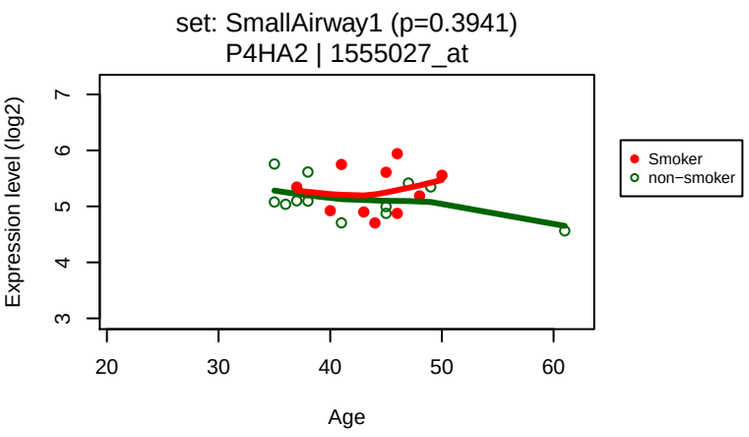
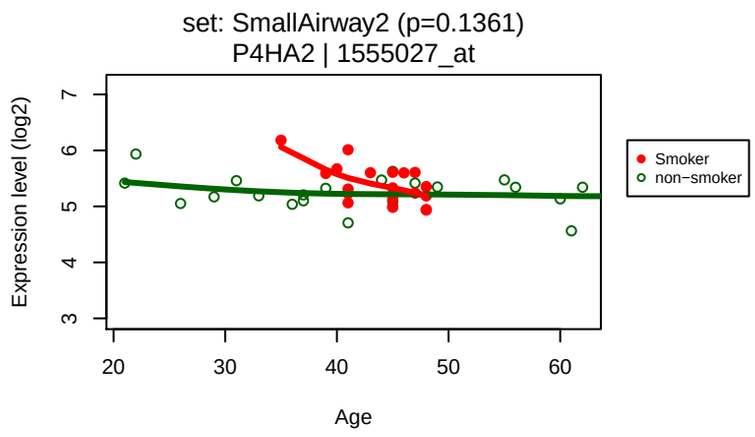
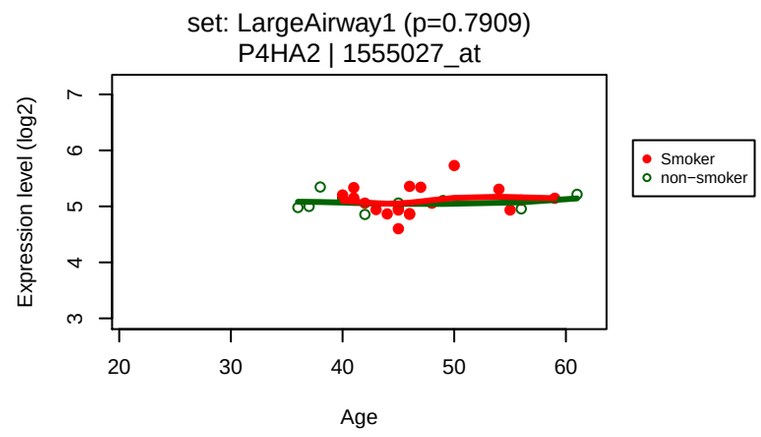

Gene expression of smokers and nonsmokers
gene: P4HA2 | 1555027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3941
p=0.1361
p=0.8302
p=0.7909
p=0.0954
p=0.8402
p=0.7993
p=0.8937
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

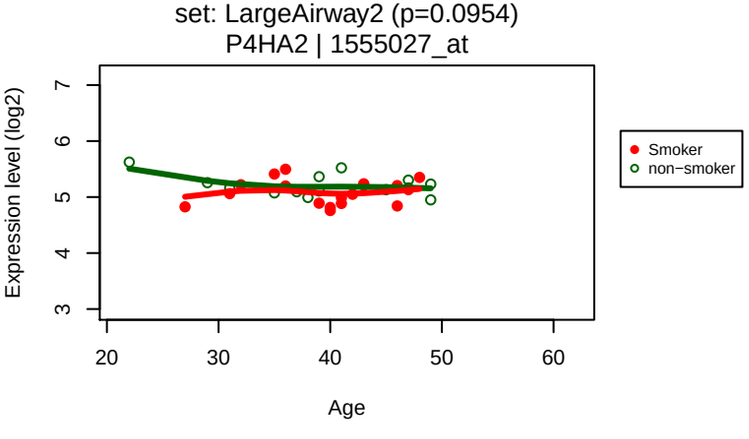

Gene expression ratio (smokers/nonsmokers)
gene: P4HA2 | 1555027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3941
p=0.1361
p=0.8302
p=0.7909
p=0.0954
p=0.8402
p=0.7993
p=0.8937
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

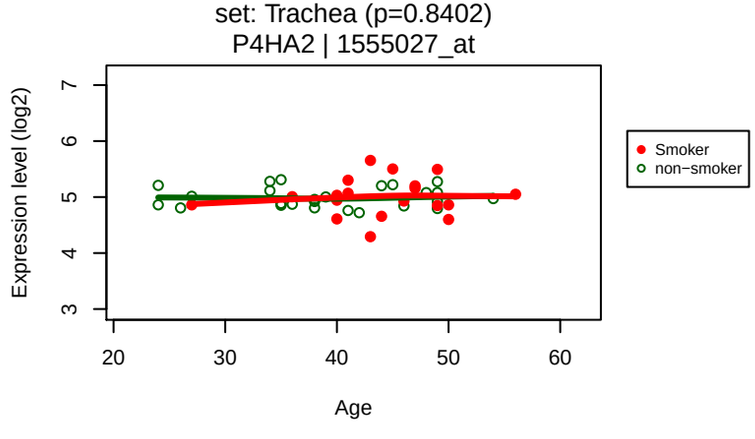

Gene expression ratio in Males
gene: P4HA2 | 1555027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0698
p=0.8737
p=0.71
p=0.8701
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P4HA2 | 1555027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9988
p=0.0379
p=0.9133
p=0.5362
SmallAirway2
LargeAirway2
Trachea
Macrophage2

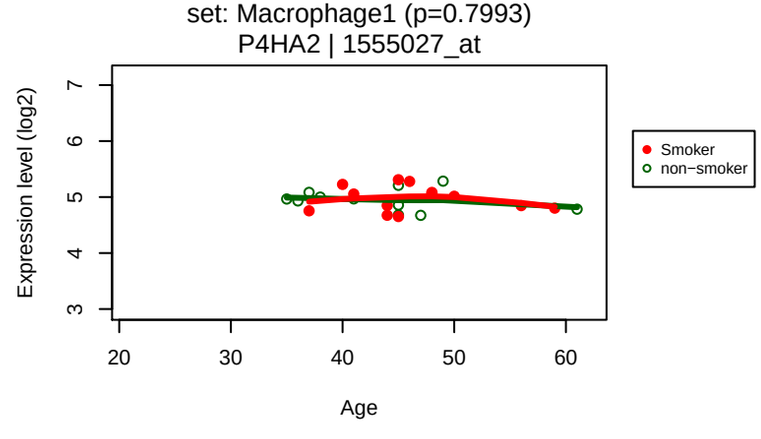

set: SmallAirway1 (p=0.3941)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker

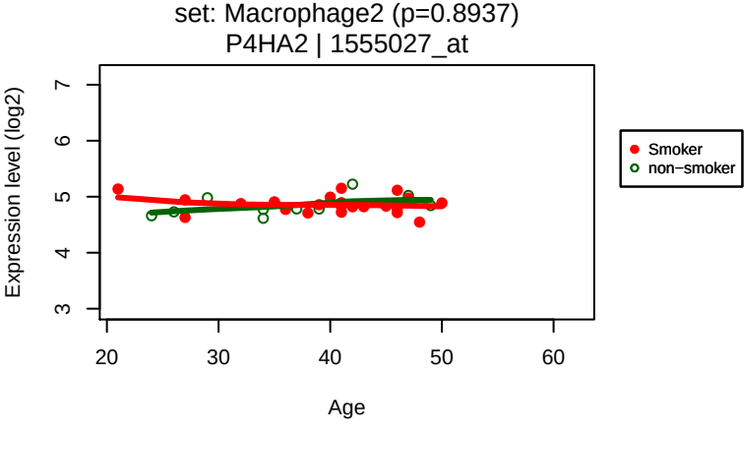

set: SmallAirway2 (p=0.1361)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker

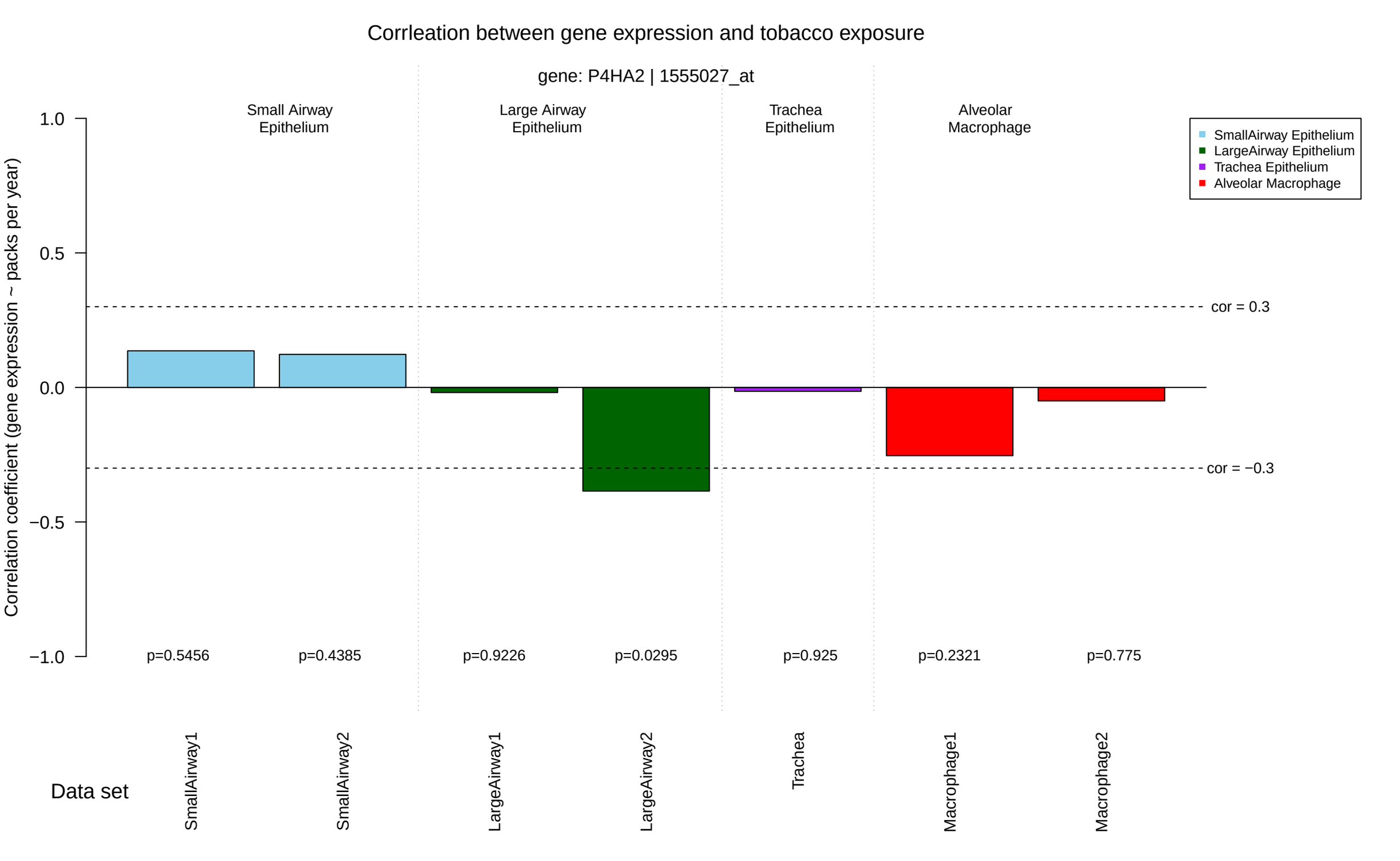

set: LargeAirway1 (p=0.7909)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0954)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8402)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7993)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8937)
P4HA2 | 1555027_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P4HA2 | 1555027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5456
p=0.4385
p=0.9226
p=0.0295
p=0.925
p=0.2321
p=0.775
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2